Team Control Number

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Summary

With growing concerns over the environmental impact of sport mega-events, the selection of a sustainable host city has become a means of minimizing overall CO_2 emissions. The Super Bowl, the most popular sporting event in the United States—ranked fifth internationally in global viewership—attracts an average of 70,000 attendees. As a result of its immense popularity, the 3-4 hour event has been estimated to produce several tons of CO_2 emissions, much of which can be attributed to the inflow of people within the host city.

We aim to formulate a comprehensive model that effectively measures the sustainability of potential host cities with respect to several key factors. Our work is divided into two major components. First, we identify and model the predominant contributors to emissions of large sporting events, with prior research showing that transportation is the greater contributor. Accordingly, we created transportation models for both travel by plane and car by using national transportation data and carbon emissions by distance, yielding a framework for estimating external transportation emissions for any city in the U.S. Second, to assess cities' sustainability and resilience, we develop an entropy-based environmental impact metric. Using Shannon entropy over eight emissions categories (residential, commercial, industrial, and transportation). A city with a higher entropy represents more balanced emissions between these categories, which further indicates better adaptability in response to the environmental impact of hosting a sport mega-event, such as the Super Bowl.

After using SQL to build a city-based dataset, we apply our dual-component model. We find that industrial and commercial gas emissions are the most influential factors relative to sustainability, while international transportation emissions (rather than inter-city transportation) seem to have a less pronounced role. Our combined model outputs the most environmentally sustainable locations for hosting the 2029 Super Bowl. In addition, we generalized our model to other sport mega-events, including Formula 1 Grand Prix and the National Senior Games, as well as rural events using a trained neural network to evaluate emissions in areas with limited environmental data.

Ultimately, our model provides a scalable and comprehensive approach to determining the most environmentally sustainable host cities for sport mega-events in light of global efforts to reduce carbon emissions.

Contents

1 Letter to the NFL **2**

2 Introduction **3**
2.1 Background . . . 3
2.2 Problem Restatement . . . 4

3 Assumptions and Variables **4**
3.1 Justifications . . . 4
3.2 Variables . . . 5
3.3 Data Collection . . . 6
3.4 Derivation . . . 6
3.4.1 D_i^{ctr}, D_i^{st} . . . 6
3.4.2 E_i . . . 7

4 Modeling **7**
4.1 External Emissions Calculation . . . 7
4.1.1 Car Emissions . . . 7
4.1.2 Plane Emissions . . . 8
4.2 City Entropy and Impact . . . 12
4.3 Comparison of Model Results . . . 13
4.4 Sensitivity Analysis . . . 14
4.4.1 Partial Derivative Equation Derivation . . . 14
4.4.2 Evaluating Partial Derivatives . . . 15
4.4.3 Analysis of Results . . . 16

5 Generalization **16**
5.1 Other Sport Mega-Events . . . 16
5.1.1 Formula 1 Grand Prix . . . 16
5.1.2 National Senior Games . . . 17
5.2 Rural Events . . . 18

6 References **19**

7 AI Appendix **20**

1 Letter to the NFL

Dear NFL Super Bowl Selection Committee,

We are writing to present our recommendation for the host city of Super Bowl LXIII in 2029, based on a comprehensive evaluation of environmental sustainability. After considering several locations using a quantitative model that accounts for intercity transportation-based emissions as well as residential, industry, commercial, and transportation emissions within a city, our team has

determined that **New Orleans, LA** (for a city scheduled to host) or **Green Bay, WI** (for the best possible historical host) would be the most environmentally responsible choice for hosting the Super Bowl. We employed a two-part model that assessed the effects of the following factors on a city's environmental sustainability: external transportation-based emissions, internal residential-based emissions, internal industry-based emissions, internal commercial-based emissions, and internal transportation-based emissions.

As the NFL continues to emphasize sustainability, choosing either of these cities for Super Bowl LXIII would be an appropriate step toward that goal. Our analysis suggests that this city is a reasonable, low-emission option that aligns well with the league's long-term environmental commitments. We look forward to seeing this approach help guide the planning of future Super Bowls.

Sincerely,
Team #17586

2 Introduction

2.1 Background

Large-scale sports events such as the Super Bowl draw millions of viewers and enable massive flows of people, resources, and infrastructure. While these events generate large commercial and economic value, their environmental footprint is equally substantial and should thereby serve as a key consideration in determining ideal locations to host them. In doing so, it is necessary to understand the full scope of the innumerable ecological impacts—greenhouse gas emissions, energy usage, waste generation, transportation emissions, among others—as environmental sustainability becomes increasingly important in planning these events. However, in practice, environmental considerations seldom take precedent in determining host cities, despite significant location-based variability in carbon footprints.

Prior research affirms that sport mega-events are immense sources of carbon emissions. One particular study that evaluates a framework for estimating these impacts demonstrates how operational waste alone can reach thousands of tons, which consequently places pressure on city waste systems. They also emphasize the energy intensity of stadiums, particularly a 55,000-seat modern venue, which typically consumes around 10,000 MWh of electricity annually, an amount sufficient to power up to 1,000 average American households for a year. [1] Another study claims that a key contributor to carbon emissions from large-scale sporting events is the flow of people within the city where the stadium is located, including the transportation of athletes and spectators, the construction of infrastructure and event venues, and the types of energy used to effectively execute the event. Relative to each other, however, they note that transportation to and from the event often dominates in emissions per capita. Most notably, spectator travel, primarily by plane and car, has been consistently identified as the largest

source of emissions, with lodging and other basic necessities playing a role as well. [2] Regarding the measurement of city sustainability, Mahmoudzadeh et al. assessed various geographic factors, including air quality, land surface temperature, vegetation coverage, and population density. Their study suggests that locations with moderate climates, better air quality, and efficient and effective waste and population management mechanisms are generally associated with higher environmental quality. [3] Ultimately, these studies show that both a city's environmental conditions and the flow of people within a city heavily characterize the carbon footprint of sports mega-events.

2.2 Problem Restatement

1. Identify the primary sources of environmental impact in hosting sports mega-event, using Super Bowl LIX in New Orleans as the baseline
2. Develop a model that evaluates the environmental sustainability of potential Super Bowl cities based on environmental considerations.
3. Apply the model to determine which location would be the most sustainable choice for the 2029 Super Bowl.
4. Expand the model to other large-scale sporting events and analyze how changes in key factors influence the model's output.

3 Assumptions and Variables

3.1 Justifications

1. **Assumption:** The environmental impact of hosting the Super Bowl at a given stadium can be attributed to interactions with its encompassing city.
    - **Justification:** Lacking data on visitor impacts in the scope of individual stadiums, we assume that the impact is instead based on their presence in a given city [1]. Based on prior research, transportation and accommodations primarily make up environmental impact of large sporting events, further backing the decision to focus on city-level optimization.
2. **Assumption:** Possible locations for the Super Bowl only include cities with stadiums, including cities that have not previously hosted the event.
    - **Justification:** Building a stadium in a given city for the event would eclipse any other environmental impact, making it irrelevant to the goal of finding the most environmentally conscious hosting location.
3. **Assumption:** For large sporting events, the number of people attending is not dependent on the perceived desirability of going to the city.

- **Justification:** With the financial and time cost of attending a large sporting event, the perceived value of the hosting location diminishes greatly for visitors, as they are solely focused on gaining value out of their entry to the event. Thus, this assumption also does not apply to smaller events.

3.2 Variables

Symbol	City Statistic
l_i^{ϕ}	The latitude.
l_i^{λ}	The longitude.
p_i	The population.
f_i	The flight count.
a_i	Total area (km^2).
s_i	Total street length (km).
$e_i^{res,e}$	Yearly residential GHG emissions from electricity (CO_2e).
$e_i^{res,g}$	Yearly residential GHG emissions from gas (CO_2e).
$e_i^{com,e}$	Yearly commercial GHG emissions from electricity (CO_2e).
$e_i^{com,g}$	Yearly commercial GHG emissions from gas (CO_2e).
$e_i^{ind,e}$	Yearly industrial GHG emissions from electricity (CO_2e).
$e_i^{ind,g}$	Yearly industrial GHG emissions from gas (CO_2e).
$e_i^{trns,g}$	Yearly vehicle GHG emissions from gas (CO_2e).
$e_i^{trns,d}$	Yearly vehicle GHG emissions from diesel (CO_2e).

Table 1: Summary of defined symbols and corresponding sourced statistics for given cities.

Symbol	City Statistic
D_i^{ctr}	Country population-weighted average distance.
D_i^{st}	State population-weighted average distance.
E_i	Yearly total GHG emissions (CO_2e).

Table 2: Summary of defined symbols and corresponding derived statistics for given cities.

Symbol	Center Statistic
L_i^{ϕ}	The latitude.
L_i^{λ}	The longitude.
P_i	The population.

Table 3: Summary of defined symbols and corresponding statistics for population centers.

Function	Description
$d(\phi_1, \lambda_1, \phi_2, \lambda_2)$	Distance between (ϕ, λ) points in km.
$D^c(\phi, \lambda)$	Country pop-weighed average distance for (ϕ, λ).
$D^s(\phi, \lambda)$	State pop-weighed average distance for (ϕ, λ).

Table 4: Summary of defined symbols and corresponding statistics for population centers.

Symbols	Source
l_i^ϕ, l_i^λ, p_i	[4] SimpleMaps US Cities Database
f_i	[5] Bureau of Transportation Statistics
a_i, s_i	[6] OSMnx Street Networks Dataverse
e_i^*, E_i	[7] OpenEI City and County Energy Profiles
L_i^ϕ, L_i^λ, P_i	[8] US Census Bureau

Table 5: Summary of key variables and their corresponding data sources.

3.3 Data Collection

The dataset used for the model is constructed by collating multiple CSV datasets using SQL queries, including cities, flights, visits, census centers, energy emissions, Super Bowl attendance, renewable-energy purchases, and transportation network data. Additional derived variables were added programmatically to the created table.

The final dataset is produced by left-joining all sources to the cities table based on either city/state names or spatial proximity. Aggregated variables include the number of flights, average Super Bowl attendance, and sum of renewable purchases. All results are grouped at the city level and filtered to cities with a population above 50,000.

3.4 Derivation

Certain higher-order variables were derived based on the core data from sources.

3.4.1 D_i^{ctr}, D_i^{st}

The population-weighed average distance for a latitude and longitude is computed in comparison to the n population clusters given by US Census Bureau data, as below.

$$D_i^{ctr}(\phi, \lambda) = \sum_{i=1}^{n} \frac{d(\phi, \lambda, L_i^\phi, L_i^\lambda) P_i}{\sum_{j=1}^{n} P_j}$$

The function $D_i^{st}(\phi, \lambda)$ is equivalent, with the modification of limiting the n population clusters to only those in the same state as the city. $d(\phi_1, \lambda_1, \phi_2, \lambda_2)$

is a function that gives the distance between two points defined in latitude and longitude.

$$d(\phi_1, \lambda_1, \phi_2, \lambda_2) = \sqrt{((\phi_2 - \phi_1) * 111.32)^2 + ((\lambda_2 - \lambda_1) * 111.32 * \cos(\frac{\pi}{360}(\phi_1 + \phi_2)))^2}$$

The population-weighted average distance for the ith city is thus computed based on $D_i^{ctr}(\phi, \lambda)$ and $D_i^{st}(\phi, \lambda)$, as follows.

$$D_i^{ctr} = D_i^{ctr}(l_i^{\phi}, l_i^{\lambda}) \quad D_i^{st} = D_i^{st}(l_i^{\phi}, l_i^{\lambda})$$

3.4.2 E_i

The total yearly GHG emissions for the ith city is simply a sum of the granular yearly GHG emissions.

$$E_i = e_i^{res,e} + e_i^{res,g} + e_i^{com,e} + e_i^{com,g} + e_i^{ind,e} + e_i^{ind,g} + e_i^{trns,g} + e_i^{trns,d}$$

4 Modeling

Our efforts in modeling the environmental impact of cities consisted of two separate yet interwoven parts. [1] The first part utilizes our preliminary research that transportation represents the greatest contributor of GHG emissions, meaning that it is of utmost importance to dedicate resources to modeling it. Thus, the first part of our model specializes on accurately computing transportation emissions per capita. For the second part, we wanted to take into consideration the overall sustainability of cities, thought of as how resilient the total E_i is to each of its cumulative parts. This sustainability score was computed using Shannon entropy, providing a quantifiable measure for how individual cities' environmental impacts respond to averse factors.

4.1 External Emissions Calculation

4.1.1 Car Emissions

An early approach to this calculation involved multiplying national transportation carbon emissions by the distance, but we found that this ignores that every route we need to calculate passes through our city, which results in a proportionally higher portion of driving through the city. We adjust for this error utilizing additional computations.
First, we will define some variables used later in this part:

$$c_{ratio} = \frac{s_i}{\sum_{j=1}^{n} s_j}$$

$$c_{rad} = \sqrt{\frac{a_i}{\pi}}$$

$$d_{car} = d(\phi, \lambda, l_i^{\phi}, l_i^{\lambda})$$

We know that the national average transportation emissions A, by definition, is a weighted average of the city and the rest of the country:

$$A = c_{ratio}c_{ghg} + (1 - c_{ratio})o_i$$

Where c_{ratio} is the proportion of roads that are in the city compared to the nation, and c_{ghg} is the transportation emissions in the city. Isolating o, the approx emissions per km in the rest of the route:

$$o = \frac{A - c_{ratio}c_{ghg}}{1 - c_{ratio}}$$

Now, we perform the correct weighting for our problem, weighting it by c_{rad}, the average radius of the city, and d_{car}, the distance that is being traveled.

$$E = (d_{car} - c_{rad}) * o + c_{rad}c_{ghg}$$

$$E = (d_{car} - c_{rad}) * \frac{A - c_{ratio}c_{ghg}}{1 - c_{ratio}} + c_{rad}c_{ghg}$$

We calculate c_{ghg} from data, where c_{km} is street length:

$$c_{ghg} = \frac{c_{emissions}}{c_{pop}c_{km}}$$

We also performed an estimation of highway distance. [9] We found in previous research that an estimation of 1.3 times the great circle distance is optimal, with an error of 0.1. Thus, $d_{car} = 1.3d$. [10] As a final step, in order to calculate per capita emissions, we use a source that averages the number of occupants in a car as 1.5.

$$E = p\frac{(1.3d - c_{rad}) * \frac{A - c_{ratio}c_{ghg}}{1 - c_{ratio}} + c_{rad}c_{ghg}}{1.5}$$

4.1.2 Plane Emissions

We estimate the additional number of business planes flying due to the Super Bowl from the pattern over the last few years. [11] We utilize data from a source, shown in the table below.

Year	**Planes**
2025	1400
2024	1700
2023	2400
2022	2000

We did not see a clear pattern in the number of business planes year-over-year, so we decided to use 1875, the average of the past 4 years, as the expected value for 2026. We also use the expectation of around 1 superbowl attendee per private jet. [12] We found a paper that found that the 2023 Super Bowl produced 1.5kt of CO_2 emissions from Super Bowl jets. By running a population-weighted sum of distance, we find that the total traveled km by 2400 private planes to Glendale is:

$$5909002.576$$

1.5 kilotons = 1.5 billion grams CO_2, so:

$$\frac{1500000000}{5909002.576} = \frac{CO_2}{km}$$

$$253.849 = \frac{CO_2}{km}$$

[13] Next, we researched a carbon emissions calculator, which has an open-source equation. For a single passenger, the calculation compromises of:

$$\frac{ax^2 + bx + c}{S * PLF} * (1 - CF) * CW * (EF * M + P) + AF * x + A$$

With the values in the following tables, for short-haul ($< 1500km$) and long-haul flights ($> 2500km$), seen below, using a linear interpolation in the middle.

Statistic	**Value**
a	0.000007
b	2.775
c	1260.608
Average Number of Seats (S)	157.86
Passenger Load Factor (PLF)	0.796
Detour Correction (DC)	95
Cargo Factor (CF)	0.26
Economy Class (CW)	1
Business/First Class (CW)	1.5
Emissions Factor (EF)	3.16
Pre-Production (P)	0.538
Multiplier (M)	3
Aircraft Factor (AF)	0.00034
Airport/Infrastructure (A)	11.68

Table 6: Values for short-haul flights.

Statistic	Value
a	0.00029
b	3.475
c	3259.691
Average Number of Seats (S)	302.58
Passenger Load Factor (PLF)	0.82
Detour Correction (DC)	95
Cargo Factor (CF)	0.26
Economy Class (CW)	1
Premium Economy Class (CW)	1.5
Business Class (CW)	4
First Class (CW)	5
Emissions Factor (EF)	3.16
Pre-Production (P)	0.538
Multiplier (M)	3
Aircraft Factor (AF)	0.00034
Airport/Infrastructure (A)	11.68

Table 7: Values for long-haul flights.

[14] According to CAPA, over the top 50 carriers, 83% of seats are economy class, 5% are economy-plus, 12% are business, and 0.6% are first class, on long-haul. On all aircraft, it's 82%, 8%, 9%, and 0.3%. Long-haul aircraft represent 19.4% of the 35,550 aircraft as of June 2025, so through computations, we get approximately 6896.7 long-haul aircraft, and 28,653.3 short and medium-haul (which will be short haul for simplicity). Long-haul aircraft have an average of 302.58 seats, so total seats:

$$6896.7 * 302.58 = 2086803.486$$

while short-haul aircraft have an average of 157.86 seats, so total seats:

$$28653.3 * 157.86 = 4523209.9$$

Doing some math, we get to that short-haul flights have this cabin ratio:

$$(82.612, 9.507, 7.716, 0.163)$$

in terms of percentages.
Since $S * PLF$ gives expected passengers on the plane, we can account for most of the sum very easily, by multiplying by $S * PLF$. For CW, we calculate it proportionally. For short-haul:

$$\frac{82.612 * 1 + 9.507 * 1 + 7.716 * 1.5 + 0.163 * 1.5}{100} = 1.039$$

And for long haul:

$$\frac{82 * 1 + 8 * 1.5 + 9 * 4 + 0.3 * 5}{100} = 1.315$$

The other terms are trivial to sum. We use CF even though we account for all emissions because the new planes added to the Super Bowl can lighten the cargo load for each plane, reducing emissions. Additionally, DC is to account for the fact that flights often experience deviations from perfect great circle distance, with the calculator accounting for this by adding 95 km. We assume that the number of new jets is equal to the number of new people divided by the average amount of passengers per plane. [15] Since according to a source, Super Bowl cargo was up 38% from normal levels, 24% of the equations below are related to cargo; when cargo is increased by 1.38 we multiply by an extra $(24 * 1.38 - 24)$ % to the equation. Short-haul: $(x < 1500)$

$$\frac{10.018 * 1.0394 * 1.0912 * (0.000007km^2 + 2.775km + 1260.608)}{125.656} + 0.00034km + 11.68$$

Long-haul: $(x > 2500)$

$$\frac{10.018 * 1.315 * 1.0912(0.00029km^2 + 3.475km + 3259.691)}{248.116} + 0.00034km + 11.68$$

Linear interpolation for $1500 \leq km \leq 2500$: the number of passengers is equal to:

$$\frac{248.116 - 125.656}{1000} * (km - 1595) + 125.656$$

And emissions per jet:

$$141.324 * (km - 1595) + 66352.795$$

So the full equation is equal to (1500-2500)

$$1000(\frac{141.324 * (km - 1595) + 66352.795)}{248.116 - 125.656 * (km - 1595) + 125.656})$$

And that gives us our full plane model $E(km)_p$, where p is people:

$$p \begin{cases} 0 & x \leq 95 \\ \frac{10.018*1.0394*1.0912*(0.000007km^2+2.775km+1260.608)}{125.656} + 0.00034km + 11.68 & x \leq 1595 \\ \frac{1000(141.324*(km-1595)+66352.795))}{248.116-125.656*(km-1595)+125.656} & 1595 < x < 2595 \\ \frac{10.018*1.315*1.0912(0.00029km^2+3.475km+3259.691)}{248.116} + 0.00034km + 11.68 & 2595 \leq x \end{cases}$$

Remember that we should input $d + 95$ into this function.

Now, we model whether people would take a car or plane over long distances. [16] According to a source, use of air travel becomes more frequent as trip distance gets longer when traveling:

Miles	% Plane
50-99	3%
100-249	6%
250-499	33%
500-749	64%
1500+	90%

Table 8: Use of personal vehicle and air travel over distance ranges.

We then performed a least-squares logistic regression on the values to arrive at a probability formula.

$$vs = \{(75, 3), (175, 6), (375, 33), (625, 64), (1500, 90)\}$$
$$y = \frac{89.31122}{1 + e^{-(0.0043793x - 3.36235)}}$$

x is the distance in km, and y is the probability of passengers taking a flight instead of a car (in %), thereby computing the distribution of the main transportation methods.
We then take a weighted average to get our final equation for emissions. If c is the chance of air travel:

$$E(d)_t = (p - 1875)(c * E(km + 95)_p + (1 - c)E(km)_c) + 1875 * 253.82d$$

And as cited earlier, we use $p = 77455.94231$. Giving us our final model for total emissions:

$$E(d)_t = 75580.94231(c * E(d + 95)_p + (1 - c)E(d)_c) + 1875 * 253.82d$$

And per capita:

$$E(d)_t = \frac{75580.94231(c * E(d + 95)_p + (1 - c)E(d)_c) + 1875 * 253.82d}{77455.94231}$$

4.2 City Entropy and Impact

As we mentioned earlier, for the second part of the model we used Shannon entropy H to model how sustainable or resilient a city is. The goal of this part was to find a numeric value for the amount of entropy of all cities and then to find the plausible city that has the highest entropy because that represents the most sustainable or resilient city.
We start with 4 sections of data (8 categories total): residential (electric and gas emissions), commercial (electric and gas emissions), industry (electric and gas emissions), and transportation (electric and diesel).
For quantifying the sustainability of a given city, we utilize Shannon entropy over E_i and its comprising metrics:

$$E_i^x = \{e_i^{res,e}, e_i^{res,g}, e_i^{com,e}, e_i^{com,g}, e_i^{ind,e}, e_i^{ind,g}, e_i^{trns,g}, e_i^{trns,d}\}.$$

For Shannon entropy, we use this equation, where $i = 1, 2, ..., 8$ and $n = 8$:

$$-\sum_{i=1}^{n} p_i \log_2 p_i$$

After calculating S_i for each ith city, we select the city with the highest Shannon entropy. In this case, a higher entropy indicates greater uncertainty in predicting the emissions category contributing most, meaning that emissions are more evenly distributed across categories. A balanced environmental profile corresponds to diversified practices rather than heavy investment into a single high-emission sector, signaling that a city can accommodate an increased Super Bowl population in an environmentally conscious manner.
In order to convert a list of GHG emissions (our 8 categories) into a probability distribution for Shannon entropy, we utilized a safe softmax function. First, we defined $m = \max(E_i \ \forall i)$. For each of the n categories in E_i, we calculate probability as:

$$w_i = \frac{e^{(E_i^x - m)*10^{-6}}}{\sum_{j=1}^{n} e^{(E_j^x - m)*10^{-6}}}$$

Lastly, after computing this probability for each ith city p_i, we plug it into the Shannon entropy equation to get the resultant entropy of each function.

4.3 Comparison of Model Results

We ran and calculated both parts of the model on the candidates for Super Bowl 2029 (external transportation-based emissions and internal entropy). Our analysis suggests that Green Bay, WI would be the most environmentally sustainable location to host the Super Bowl, while New Orleans, LA ranked the highest out of the cities scheduled to host in the future.

City	**Emissions Ranking**	**Entropy Ranking**	**Mean Ranking**
Green Bay, WI	9	1	5
Cincinnati, OH	5	7	6
Cleveland, OH	10	5	7.5
Minneapolis, MN	11	4	7.5
Pittsburgh, PA	13	3	8
Kansas City, MO	1	16	8.5
Buffalo, NY	17	2	9.5
New Orleans, LA	14	8	11
Nashville, TN	2	22	12
Chicago, IL	4	20	12

4.4 Sensitivity Analysis

4.4.1 Partial Derivative Equation Derivation

The partial derivative of the Shannon entropy function with respect to a specific probability p_1, using log base 2 (resulting in units of bits), is given by the formula:

$$\frac{\partial H}{\partial p_1} = -(\log_2 p_1 + \frac{1}{\ln 2})$$

The Shannon entropy H for a discrete probability distribution with N possible outcomes, using base 2, is defined as:

$$H = -\sum_{i=1}^{N} p_i \log_2(p_i)$$

To differentiate this, we use the change-of-base formula to convert $\log_2(p_i)$ to the natural logarithm (ln):

$$\log_2(x) = \frac{\ln(x)}{\ln(2)}$$

Substituting this into the entropy formula:

$$H = -\sum_{i=1}^{N} p_i \frac{\ln(p_i)}{\ln(2)} = -\frac{1}{\ln(2)} \sum_{i=1}^{N} p_i \ln(p_i)$$

Here, $\frac{1}{\ln(2)}$ is a constant scaling factor.

Now, we solve for the derivative:

1. Applying the Sum Rule and Constant Multiple Rule:

$$\frac{\partial H}{\partial p_1} = -\frac{1}{\ln(2)} \frac{\partial}{\partial p_1} \left(\sum_{i=1}^{N} p_i \ln(p_i) \right)$$

2. Differentiating Each Term: The derivative of the sum simplifies to just the derivative of the p_1 term, using the product rule:

$$\frac{\partial}{\partial p_1}(p_1 \ln(p_1)) = \ln(p_1) + 1 \quad \text{for } i = 1$$

   Because we are using partial derivatives, all other terms ($i \neq 1$) differentiate to zero.

3. Combining the Results:

$$\frac{\partial H}{\partial p_1} = -\frac{1}{\ln(2)}(\ln(p_1) + 1)$$

4. **Solving for $\frac{\partial H}{\partial p_1}$:**
   Finally, we can distribute the constant and use the change of base formula ($\ln(p_1) = \ln(2) \cdot \log_2(p_1)$) to express the final answer in terms of $\log_2$:

$$\frac{\partial H}{\partial p_1} = -(\frac{\ln(2)\log_2(p_1)}{\ln(2)} + \frac{1}{\ln(2)})$$

$$\frac{\partial H}{\partial p_1} = -(\log_2 p_1 + \frac{1}{\ln 2})$$

   This result assumes the probabilities p_i are independent variables. The constant $\frac{1}{\ln 2}$ has a numerical value of approximately 1.4427.

5. Generalization: Finally, we generalize this for all p_i:

$$\frac{\partial H}{\partial p_i} = -(\log_2 p_i + \frac{1}{\ln 2})$$

4.4.2 Evaluating Partial Derivatives

After calculating the partial derivatives of each category and averaging all the values for each city, we get a list of the level of importance of each variable. In other words, the higher the partial derivative for a variable, the more changing that variable affects the sustainability score. From our analytical calculations, after applying softmax ($S_i = \frac{e^{s_i}}{\sum_{j=1}^{n} e^{s_j}}$), we get this table of values:

Category (in CO_2e)	**Partial ∂**
Industry gas	0.232
Commercial gas	0.229
Industry electric	0.180
Transport diesel	0.122
Residential gas	0.112
Residential electric	0.069
Commercial electric	0.054
Transport gas	0.002

We can also combine the mediums for each metric to get the total impact caused by each category.

Category (in CO_2e)	**Partial ∂**
Industry	0.672
Commercial	0.201
Residential	0.124
Transport	0.003

4.4.3 Analysis of Results

These results line up with scientific expectations. Industry emissions (both gas and electric emissions) impact sustainability more than any other category. In addition, transport (which accounts for transport within a city and not external transport by air) is barely impactful environmentally. It also seems that gas sustainability should be prioritized as a whole across cities.

5 Generalization

5.1 Other Sport Mega-Events

5.1.1 Formula 1 Grand Prix

We now use our generalized model and apply it to the Formula 1 Grand Prix tournament. To do so, we utilize the same entropy model as before and find the same type of data about the categories for all eligible cities: namely, all cities that have big enough racing tracks. Although other cities have hosted in the past, currently only 3 cities host the Grand Prix: Austin, TX, Miami, FL, and Las Vegas, NV. The model output is shown in the following table.

City	**Emissions Ranking**	**Entropy Ranking**	**Mean Ranking**
Sebring, FL	3	1	2
Indianapolis, IN	1	5	3
Miami, FL	4	3	3.5

The results of the entropy model showed that the optimal location for the Grand Prix to minimize the environmental impact based on the sustainability of the city is **Sebring, FL**. Thus, Sebring would be the best choice considering all cities that have hosted or plan to host. However, of the cities that currently host, the best choice would be **Miami, FL**, which is not far behind Sebring in terms of sustainability.

Category (in CO_2e)	**Partial ∂**
Industry gas	0.210
Commercial gas	0.199
Residential gas	0.182
Industry electric	0.134
Transport diesel	0.111
Commercial electric	0.090
Residential electric	0.063
Transport gas	0.011

Category (in CO_2e)	**Partial ∂**
Industry	0.478
Commercial	0.305
Residential	0.196
Transport	0.020

From the sensitivity analysis results, we see that the *industry* sector, specifically *industry gas*, has the largest impact on the entropy of the considered cities. Thus, a city aiming to emphasize its sustainability for the Grand Prix should put investment into a more eco-friendly industrial sector. Additionally, based on this result and the table, we see that the impact of each category varies from the original result from the data on the NFL cities, suggesting that this is different and unique from our baseline model and its results.

5.1.2 National Senior Games

The National Senior Games is a U.S. multi-sport mega event for adults aged 50 years or older, drawing participation from all 50 states. Because the event takes place every two years, we apply our sustainability model to assess which potential host cities would minimize environmental impact. For this analysis, we evaluated 18 cities that have previously hosted the National Senior Games and gathered data on the same set of sustainability factors used in the baseline model. After organizing the data and running the model, each city was assigned a ranking based on their level of sustainability.

City	**Emissions Ranking**	**Entropy Ranking**	**Mean Ranking**
Des Moines, IA	2	4	3
Bloomington, MN	5	3	4
St. Paul, MN	7	5	6

After taking the arithmetic mean of the resulting rankings, the city was ranked the highest was **Des Moines, IA**. Furthermore, after applying the same softmax from previous sensitivity calculations, we get:

Category (in CO_2e)	**Partial ∂**
Commercial gas	0.228
Residential gas	0.194
Industry gas	0.150
Industry electric	0.131
Transport diesel	0.111
Commercial electric	0.088
Residential electric	0.088
Transport gas	0.011

Category (in CO_2e)	**Partial ∂**
Commercial	0.348
Industry	0.339
Residential	0.293
Transport	0.020

From the sensitivity analysis results, we see that the *commercial* sector, specifically *commercial gas*, has the largest impact on the entropy of the considered cities. Thus, a city aiming to emphasize its sustainability for the National Senior Games should put focus on developing a more environmentally conscious commercial sector. Additionally, on this result and the table, we see that the impact of each category varies from the original result from the data on the NFL cities, suggesting that this is different and unique from our baseline model and its results.

5.2 Rural Events

Some sporting events are hosted in rural areas away from the cities, which means we lack the basis of internal emissions which is used in the previous models. For computing the GHG emissions per capita within these areas, a neural network is employed, defined as follows.

$$f : \mathbb{R}^{19} \to \mathbb{R} \quad x : \mathbb{R}^3$$

$$f(x) = W^{(3)}\sigma(W^{(2)}\sigma(W^{(1)}x + b^{(1)}) + b^{(2)}) + b^{(3)}$$

$$W^{(1)} \in \mathbb{R}^{16\times 19}, W^{(2)} \in \mathbb{R}^{16\times 16}, W^{(3)} \in \mathbb{R}^{1\times 16}$$

Training consists of gradient-based optimization using ϕ and λ, computing loss based on known $\frac{E_i}{p}$ (CO_2e per capita) values from n cities. [ai/2] The encode function, $\mathcal{E}(x)$, adds non-linear spatial dimensionality, as $\mathbb{R}^3 \to \mathbb{R}^{19}$. The loss function, $\mathcal{L}(x)$, computes root mean squared error (RMSE).

$$\theta = \{W^{(1)}, b^{(1)}, W^{(2)}, b^{(2)}, W^{(3)}, b^{(3)}\}$$

$$\min_{\theta} \sum_{i=1}^{n} (f(\mathcal{E}(x_i); \theta) - \frac{E_i}{p_i})^2$$

$$\theta_{t+1} = \theta_t - \nabla_\theta \mathcal{L}(\theta_t) \text{ for } t = 0, \ldots, 2499$$

After training, the optimized f can be used to calculate GHG emissions for any latitude ϕ and longitude λ in the United States. As an additional requisite, the state population-weighed sum is also calculated for the point.

$$x = \begin{bmatrix} \phi \\ \lambda \\ D_i^{st}(\phi, \lambda) \end{bmatrix} \quad E = f(\mathcal{E}(x))$$

For use in planning a California event, a version of this model was trained solely on cities in and surrounding California. Then, it was sampled with $\phi \in \{32.5, 32.5{+}0.25, \ldots, 42.0\}$ and $\lambda \in \{-124.4, -124.4{+}0.25, \ldots, -114.1\}$, netting the following results for the least and most environmentally friendly coordinates.

ϕ	λ	$\frac{\hat{E}}{p}$
36.0	-120.15	6.42
33.75	-118.15	9.95

Thus, for a sport that does not require a dedicated facility, $36.0°N\ -120.15°W$ would be recommended to minimize the predicted GHG impact.

6 References

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7 AI Appendix

1. **Query**: nonlinear ghg emissions over long distances calculator

   - **Answer**: A suitable option for long-distance, nonlinear GHG estimation is the myclimate flight calculator. It incorporates great-circle routing, radiative forcing factors, aircraft class, and load factors, all of which introduce nonlinear scaling of emissions with distance rather than a simple linear multiplier.
     https://www.myclimate.org/en/information/about-myclimate/downloads/flight-emission-calculator

2. **Query**: how do I use the range function in duckdb to go from some value x to y

   - **Answer**: SELECT * FROM range(x, y);
     This yields integers from x to y-1. Add a step if needed:
     SELECT * FROM range(x, y, step);

3. **Query**: Turn a latitude and longitude to a higher dimension using sin and cos nonlinearity?

   - **Answer**: Use a multi-frequency sinusoidal encoding:
     - Choose frequencies ω_k (e.g., 2^k or a linear grid).
     - For each k, map lat ϕ and lon λ: $e_k = [\sin(\omega_k\phi),\ \cos(\omega_k\phi),\ \sin(\omega_k\lambda),\ \cos(\omega_k\lambda)]$.
     - Concatenate e_k for all k.
     - Optionally append the raw coordinates or apply different frequency sets per coordinate.

     This expands 2-D coordinates into a smooth, high-dimensional nonlinear space.

4. **Query**: what is a good dataset for city road length for US cities?

   - **Answer**: A reliable and well-structured option is Geoff Boeing's U.S. Street Network Models on Harvard Dataverse. It provides street-network data for thousands of U.S. cities, including detailed edge lists you can use to calculate total road length. The datasets are available in multiple formats (GraphML, shapefiles, CSV) and are

already cleaned and standardized, which makes analysis much easier than starting from raw TIGER/Line files.
Link: https://doi.org/10.7910/DVN/CUWWYJ